# Web01 Incident Analysis

Francisca Tupkeyi | Incident date March 12 2026

This analysis correlates the packet capture, web access and error logs, and Zeek connection, DNS and TLS records. The evidence supports a possible compromise of web01 but does not confirm malicious code execution or lateral movement.

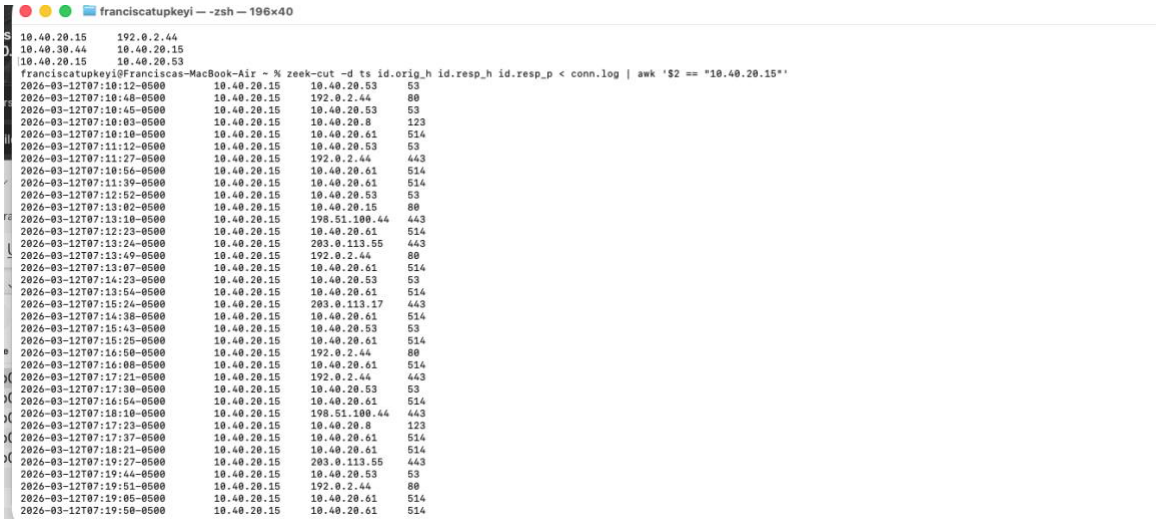
Time convention: All event times below use UTC−05:00 to match the reviewed Zeek output. For comparison with the supplied access log in UTC−04:00, add one hour. For example, 07:12:44−0500 and 08:12:44−0400 represent the same instant. The exact alert-generation time was not supplied; the first suspicious connection is the reference point.

## 1 Which host initiated the suspicious traffic

web01, with IP address 10.40.20.15, initiated outbound TCP connections to 198.51.100.44 on port 443 at 07:13:10, 07:18:10 and 07:23:10. The source is listed as id.orig_h in conn.log. The connections began exactly five minutes apart. In the PCAP, initial SYN packets 384, 1003 and 1574 independently identify web01 as the initiator. This establishes the origin of the traffic without proving that it was malicious.

Evidence: Zeek conn.log and the PCAP. Each session contains 612 bytes of outbound TCP payload and 924 bytes inbound. These totals include TLS traffic and do not establish stolen-data volume.

**Screenshot**



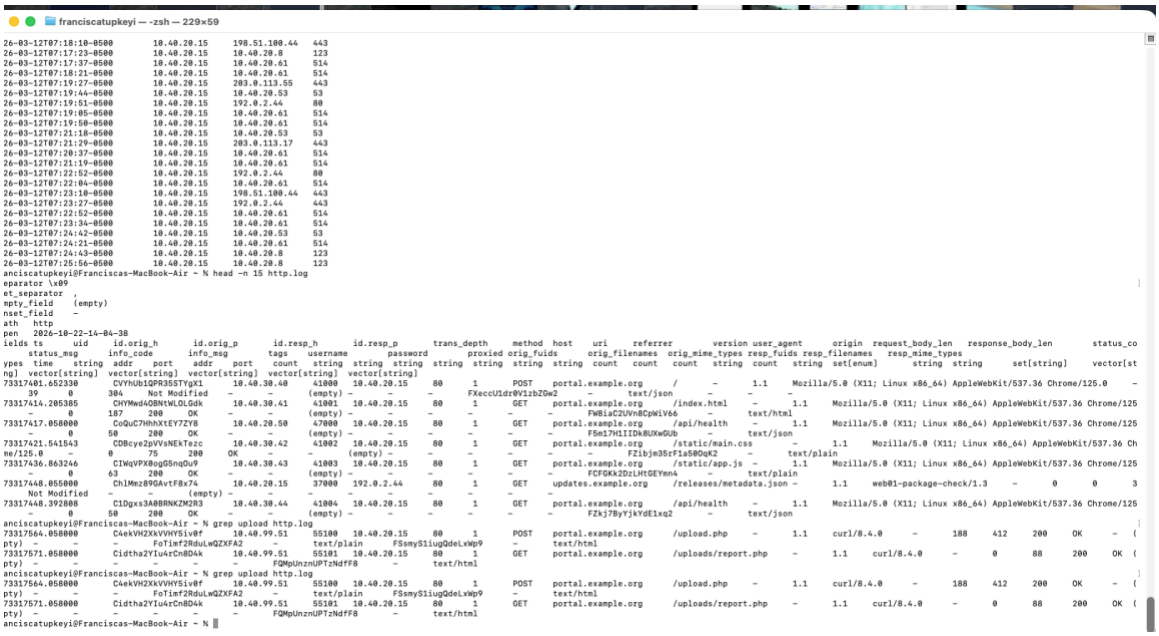

*conn.log output showing web01, 198.51.100.44, port 443 and the repeated timestamps.*

**2 What happened immediately before the alert**

At 07:12:44, internal host 10.40.99.51 sent POST /upload.php to web01. At 07:12:51, the same host requested GET /uploads/report.php. Both requests received HTTP 200 responses. The first suspicious outbound connection began at 07:13:10, which was 26 seconds after the POST and 19 seconds after the GET. This sequence is consistent with a possible malicious upload followed by access to a web shell. However, the paths, timing and response codes do not prove that a malicious file executed.

Evidence: Zeek http.log, web01_access_2026-03-12.log and PCAP request packets 329 and 340. The supplied error log contains missing favicon.ico messages and does not confirm a PHP execution error or compromise.

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**Screenshot**




*the HTTP output showing both requests, source 10.40.99.51 and HTTP status 200.*

**3 Which external systems appear in the records**

DNS and TLS records show matching names for several external destinations. Their names and behavior are consistent with background services, although names alone do not prove legitimacy.

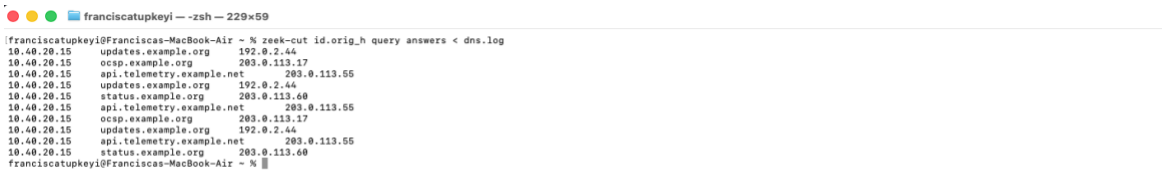
| External IP | Name | Observed evidence |
|---|---|---|
| 192.0.2.44 | updates.example.org | DNS and TLS; ports 80 and 443 |
| 203.0.113.55 | api.telemetry.example.net | DNS and TLS; port 443 |
| 203.0.113.17 | ocsp.example.org | DNS and TLS; port 443 |
| 203.0.113.60 | status.example.org | DNS answer; no matching TCP session identified |
| 198.51.100.44 | cdn-asset-sync.example | TLS and repeated port 443 sessions; no matching DNS lookup observed |

The primary suspicious destination is 198.51.100.44. Its five-minute connections appear in conn.log and its server name appears in ssl.log. No corresponding lookup appears in the captured DNS records. A missing lookup alone is not proof of malicious activity because a name could have been cached or resolved before capture. Combined with the preceding upload, the recurring traffic warrants investigation as possible command and control.

Evidence: Zeek dns.log, ssl.log and conn.log. TLS ClientHello packets 387, 1006 and 1577 contain cdn-asset-sync.example. Encrypted application content cannot be established from these records.

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**3A**




**3B**




*DNS output and TLS server-name output.*

**4 Did web01 contact internal systems afterward**

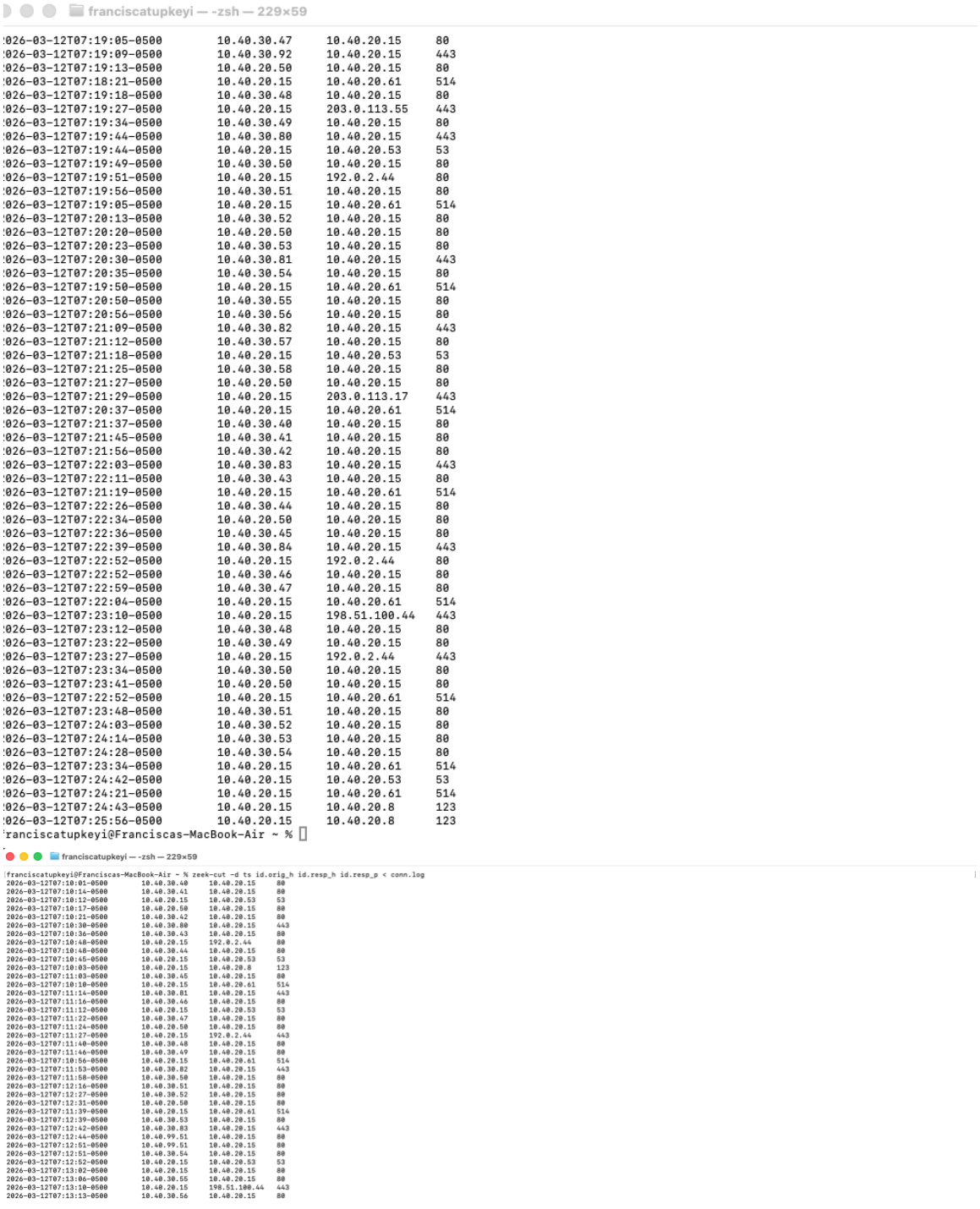
Yes. After the first suspicious connection, web01 contacted 10.40.20.61 on port 514 at 07:18:21 and 07:19:05, with further contacts through 07:24:21. The PCAP identifies this as UDP traffic consistent with syslog. Because similar traffic also exists before the suspicious sequence, it is consistent with ongoing logging rather than evidence of newly initiated lateral movement.

web01 also contacted internal DNS server 10.40.20.53 on port 53 at 07:19:44, 07:21:18 and 07:24:42. It contacted 10.40.20.8 on port 123 at 07:24:43 and 07:25:56, consistent with time synchronization. These records confirm internal communication but do not establish lateral movement. I would validate the services and compare the traffic with the normal baseline.

Evidence: Zeek conn.log and the PCAP. Requests from internal clients to web01 are inbound traffic and should be distinguished from connections initiated by web01.

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**Screenshot 4**




*the connection output showing internal destinations 10.40.20.61, 10.40.20.53 and 10.40.20.8 after 07:13:10.*

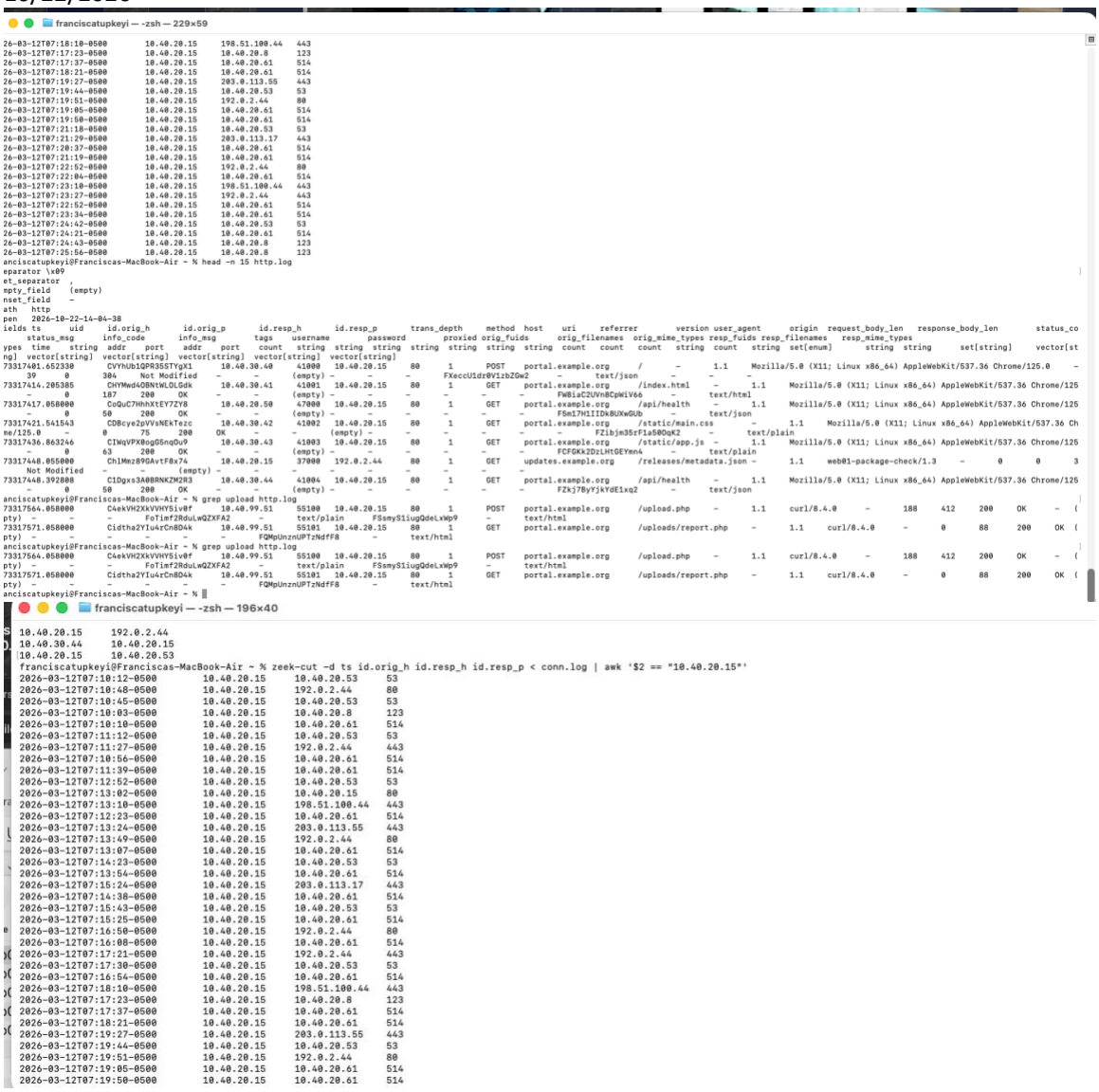
**5 What time window contains the strongest evidence**

The strongest evidence window is March 12, 2026, from 07:12:44 to 07:23:10 UTC−05:00. It includes the suspicious HTTP sequence and the start of all three repeated external connections.

| Time UTC−05:00 | Event |
|---|---|
| 07:12:44 | 10.40.99.51 sends POST /upload.php; HTTP 200 |
| 07:12:51 | 10.40.99.51 requests GET /uploads/report.php; HTTP 200 |
| 07:13:10 | web01 initiates connection to 198.51.100.44:443 |
| 07:18:10 | Second connection to the same external destination |
| 07:23:10 | Third connection continues the five-minute pattern |

The 26-second gap between the POST and first external connection creates a close temporal association. The two later connections demonstrate recurrence. This supports investigating whether the upload and outbound sessions are related, but timing alone does not establish causation.

Evidence: The HTTP timestamps in http.log and the access log, correlated with the connection timestamps in conn.log and the TLS name in ssl.log.




## Conclusion

The available evidence supports a possible compromise of web01. An internal client uploaded content and requested a PHP resource shortly before web01 began repeated encrypted connections to 198.51.100.44. These findings are consistent with possible web shell activity and command and control, but neither is confirmed. The records do not establish data exfiltration or lateral movement. I would preserve and inspect report.php, identify the process responsible for the outbound sessions, review PHP and endpoint logs,

investigate 10.40.99.51, and verify whether the external destination and internal services

are authorized.